Analysis and Design of Algorithms: Comprehensive Guide to Complexity and Paradigms

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Preface

This book provides comprehensive coverage of algorithm design and analysis, from fundamental concepts to advanced techniques. It is suitable for upper-level undergraduate and graduate computer science students studying the theory and practice of algorithms. The material emphasizes both the mathematical foundations of algorithm analysis and the practical paradigms used to design efficient solutions to computational problems.

Throughout this book, we develop intuition for algorithmic thinking, learn to analyze computational complexity rigorously, and explore design paradigms that solve diverse problems efficiently.

Contents

1 Module 1: Introduction and Algorithm Analysis **1**
1.1 What is an Algorithm 1
1.2 Pseudocode Conventions and Flowcharts 1
1.3 Time Complexity: Best, Average, Worst Case . . . 2
1.4 Asymptotic Notations 2
1.5 Common Complexity Classes 3
1.6 Space Complexity Analysis 3
1.7 Recurrence Relations 3

2 Module 2: Sorting Algorithms **7**
2.1 Bubble Sort 7
2.2 Selection Sort 7
2.3 Insertion Sort 8
2.4 Merge Sort 8
2.5 Quick Sort 9
2.6 Heap Sort 9
2.7 Counting Sort and Radix Sort 10
2.8 Sorting Algorithm Comparison 10

3 Module 3: Algorithm Design Paradigms **13**
3.1 Divide and Conquer 13
3.2 Dynamic Programming 13
3.3 Greedy Algorithms 15
3.4 Backtracking 16
3.5 Branch and Bound 16

4 Module 4: Graph Algorithms and NP-Completeness **19**
4.1 Graph Representations 19
4.2 Breadth-First Search (BFS) 19
4.3 Depth-First Search (DFS) 20
4.4 Dijkstra's Shortest Path 20

4.5 Bellman-Ford Algorithm 20
4.6 All-Pairs Shortest Paths: Floyd-Warshall 21
4.7 Minimum Spanning Tree 21
4.8 NP-Completeness 22
4.9 Approximation Algorithms 22

Module 1: Introduction and Algorithm Analysis

1.1 What is an Algorithm

An algorithm is a finite sequence of well-defined instructions for solving a computational problem. Formally, an algorithm must possess the following properties:

Finiteness: The algorithm must terminate after a finite number of steps for any valid input. An infinite loop violates this property.

Definiteness: Each instruction must be precisely defined. Ambiguous or vague instructions are unacceptable. Readers should interpret instructions identically.

Input: The algorithm receives zero or more inputs, which may come from a specified set. Well-defined input is crucial.

Output: The algorithm produces at least one output. Output must relate to inputs according to the problem specification.

Effectiveness: Every instruction must be primitive and implementable in a reasonable time. Operations must be feasible on actual computers.

These properties distinguish algorithms from mere strategies or heuristics that may violate finiteness or definiteness.

1.2 Pseudocode Conventions and Flowcharts

Pseudocode provides a language intermediate between natural language and programming languages. A pseudocode algorithm for binary search exemplifies conventions:

```
Algorithm BinarySearch(A, target, low, high)
  while low <= high do
```

```
    mid = floor((low + high) / 2)
    if A[mid] == target then
      return mid
    else if A[mid] < target then
      low = mid + 1
    else
      high = mid - 1
  return -1
```

Flowcharts visualize algorithm flow using boxes (operations), diamonds (decisions), and arrows (flow). They aid in understanding algorithm structure and verifying correctness, though they scale poorly to large algorithms.

1.3 Time Complexity: Best, Average, Worst Case

For a given input size n, an algorithm may take different times depending on input structure:

Best case: Minimum time over all inputs of size n. For linear search, best case is $O(1)$ (target at first position).

Worst case: Maximum time over all inputs of size n. For linear search, worst case is $O(n)$ (target absent or at last position). Worst-case analysis guarantees performance for any input.

Average case: Expected time over random inputs of size n. For linear search with target uniformly distributed, average case is $O(n/2) = O(n)$. Average-case analysis is harder to compute but more realistic.

1.4 Asymptotic Notations

Big-O notation characterizes algorithm growth without constant factors or lower-order terms.

Definition (Big-O): $f(n) = O(g(n))$ if there exist constants $c > 0$ and $n_0 > 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$.

Example: $f(n) = 5n^2 + 3n + 2 = O(n^2)$ with $c = 6, n_0 = 1$.

Definition (Big-Omega): $f(n) = \Omega(g(n))$ if there exist constants $c > 0$ and $n_0 > 0$ such that $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$.

Definition (Big-Theta): $f(n) = \Theta(g(n))$ if $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$. Tight bound.

1.5 Common Complexity Classes

Notation	**Name**	**Operations for** $n = 10^6$
$O(1)$	Constant	1
$O(\log n)$	Logarithmic	≈ 20
$O(n)$	Linear	10^6
$O(n \log n)$	Linearithmic	2×10^7
$O(n^2)$	Quadratic	10^{12}
$O(n^3)$	Cubic	10^{18}
$O(2^n)$	Exponential	2^{10^6}

Table 1.1: Common Complexity Classes

Algorithms with polynomial time ($O(n^k)$) are considered feasible. Exponential algorithms are infeasible for large n.

1.6 Space Complexity Analysis

Space complexity analyzes memory usage during algorithm execution. For example, merge sort requires $O(n)$ auxiliary space for merging, while quicksort requires $O(\log n)$ space for recursion.

Space and time tradeoffs are common: using more memory can reduce time (hash tables for $O(1)$ lookup vs. $O(\log n)$ binary search).

1.7 Recurrence Relations

Many algorithms have recursive structure. Recurrences analyze their complexity.

Substitution Method: Guess the solution form and verify by induction.

For $T(n) = 2T(n/2) + n$: Guess $T(n) = O(n \log n)$. Base: $T(1) = 1 = O(1)$. Step: Assume $T(n/2) = c(n/2)\log(n/2)$. Then $T(n) = 2 \cdot c(n/2)\log(n/2) + n = cn\log(n/2) + n = cn\log n - cn + n = cn\log n$ for c large enough.

Iteration/Unrolling Method: Unfold recursion and sum costs at each level.

For $T(n) = T(n-1)+n$: Unfold: $T(n) = n+T(n-1) = n+(n-1)+T(n-2) = \cdots = n+(n-1)+\cdots+1 = \frac{n(n+1)}{2} = \Theta(n^2)$.

Master Theorem: For recurrences $T(n) = aT(n/b) + f(n)$ where $a \geq 1, b > 1$, let $c = \log_b a$.

Case 1: If $f(n) = O(n^{c-\epsilon})$ for $\epsilon > 0$, then $T(n) = \Theta(n^c)$.

Case 2: If $f(n) = \Theta(n^c \log^k n)$ for $k \geq 0$, then $T(n) = \Theta(n^c \log^{k+1} n)$.

Case 3: If $f(n) = \Omega(n^{c+\epsilon})$ for $\epsilon > 0$ and $af(n/b) \leq cf(n)$ for some $c < 1$, then $T(n) = \Theta(f(n))$.

Worked Example 1: Merge Sort Recurrence

$T(n) = 2T(n/2) + n$ with $a = 2, b = 2, f(n) = n$.

$c = \log_2 2 = 1$. $f(n) = n = \Theta(n^1)$, matching Case 2 with $k = 0$.

Thus $T(n) = \Theta(n \log n)$.

Worked Example 2: Insertion Sort Recurrence

$T(n) = T(n-1) + n$ with $T(1) = 1$.

Unrolling: $T(n) = n + (n-1) + \cdots + 1 = \frac{n(n+1)}{2} = \Theta(n^2)$.

Worked Example 3: Divide and Conquer

Strassen matrix multiplication: $T(n) = 7T(n/2) + \Theta(n^2)$ with $a = 7, b = 2, f(n) = n^2$.

$c = \log_2 7 \approx 2.807$. $f(n) = n^2 = O(n^{2.807-\epsilon})$ for small ϵ, matching Case 1.

Thus $T(n) = \Theta(n^{\log_2 7}) \approx \Theta(n^{2.807})$, faster than standard $\Theta(n^3)$.

Review Questions

1. Define finiteness, definiteness, and effectiveness for algorithms.
2. Distinguish best, average, and worst-case complexity with an example.
3. Formally state the Big-O definition and give an example of $f(n) = O(g(n))$.
4. When is the Master Theorem applicable, and what are its three cases?
5. Analyze the recurrence $T(n) = 3T(n/2) + n$ using the Master Theorem.
6. What is the difference between $O(n^2)$ and $\Theta(n^2)$ complexity?
7. For an algorithm with $O(n^2)$ worst-case and $O(n)$ best-case, what can we say about average-case?
8. How would you analyze the space complexity of a recursive algorithm?

Module 2: Sorting Algorithms

2.1 Bubble Sort

Bubble sort repeatedly swaps adjacent elements if they are in wrong order.

```
Algorithm BubbleSort(A)
  n = length(A)
  for i = 1 to n-1 do
    for j = 1 to n-i do
      if A[j] > A[j+1] then
        swap(A[j], A[j+1])
```

Time complexity: Worst case $O(n^2)$ (reverse sorted array requires all swaps). Best case $O(n)$ with optimization (early termination if no swaps occur). Space: $O(1)$ in-place. Stable: Yes, equal elements maintain order.

2.2 Selection Sort

Selection sort finds minimum in unsorted portion and places at current position.

```
Algorithm SelectionSort(A)
  n = length(A)
  for i = 1 to n-1 do
    minIdx = i
    for j = i+1 to n do
      if A[j] < A[minIdx] then
        minIdx = j
    swap(A[i], A[minIdx])
```

Time complexity: $O(n^2)$ in all cases (always scans unsorted portion). Space: $O(1)$. Stable: No, swapping may disrupt order.

2.3 Insertion Sort

Insertion sort maintains sorted prefix and inserts each element into its correct position.

```
Algorithm InsertionSort(A)
  for i = 2 to length(A) do
    key = A[i]
    j = i - 1
    while j >= 1 and A[j] > key do
      A[j+1] = A[j]
      j = j - 1
    A[j+1] = key
```

Time complexity: Best case $O(n)$ (nearly sorted array). Worst case $O(n^2)$ (reverse sorted). Average $O(n^2)$. Space: $O(1)$. Stable: Yes.

2.4 Merge Sort

Merge sort divides array, recursively sorts halves, and merges.

```
Algorithm MergeSort(A, p, r)
  if p < r then
    q = floor((p + r) / 2)
    MergeSort(A, p, q)
    MergeSort(A, q+1, r)
    Merge(A, p, q, r)
```

Recurrence: $T(n) = 2T(n/2) + O(n)$, solving to $T(n) = O(n \log n)$. Space: $O(n)$ for temporary array. Stable: Yes.

2.5 Quick Sort

Quick sort partitions array around pivot, recursively sorts partitions.

```
Algorithm QuickSort(A, p, r)
  if p < r then
    q = Partition(A, p, r)
    QuickSort(A, p, q-1)
    QuickSort(A, q+1, r)

Algorithm Partition(A, p, r)
  pivot = A[r]
  i = p - 1
  for j = p to r-1 do
    if A[j] < pivot then
      i = i + 1
      swap(A[i], A[j])
  swap(A[i+1], A[r])
  return i + 1
```

Time complexity: Best/average $O(n \log n)$. Worst case $O(n^2)$ (poor pivot selection). Space: $O(\log n)$ recursion. Stable: No (unless modified). Pivot strategies: first, last, median-of-three, random.

2.6 Heap Sort

Heap sort builds max-heap and repeatedly extracts maximum.

Heap property: Parent $\geq$ children (max-heap) or Parent $\leq$ children (min-heap).

Heapify-down: Move element down until heap property restored. $O(\log n)$ time.

Build-heap: Apply heapify-down to all non-leaf nodes bottom-up. $O(n)$ time (not $O(n \log n)$).

Time complexity: Build-heap $O(n)$, extract-max n times $O(\log n)$ each, total $O(n \log n)$. Space: $O(1)$ in-place. Stable: No.

2.7 Counting Sort and Radix Sort

Counting sort and radix sort achieve linear time by assuming input constraints.

Counting sort: Assumes integers in range $[0, k]$. Create count array, scan input incrementing counts, scan counts outputting elements. Time: $O(n + k)$. Stable: Yes (process in reverse order).

Radix sort: For large integers, sort by least-significant digit, then next digit, etc. Uses counting sort as subroutine. Time: $O(d(n+k))$ where d is number of digits. Linear time when d is small.

2.8 Sorting Algorithm Comparison

Algorithm	Best	Average	Worst	Space	Stable	In-pla
Bubble	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Yes
Selection	$O(n^2)$	$O(n^2)$	$O(n^2)$	$O(1)$	No	Yes
Insertion	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	Yes	Yes
Merge	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(n)$	Yes	No
Quick	$O(n \log n)$	$O(n \log n)$	$O(n^2)$	$O(\log n)$	No	Yes
Heap	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$	$O(1)$	No	Yes
Counting	$O(n + k)$	$O(n + k)$	$O(n + k)$	$O(k)$	Yes	No

Table 2.1: Sorting Algorithm Comparison

Merge sort and quicksort are most practical: stable sorts prefer merge, in-place prefer quick. Counting/radix excel for small digit range. Insertion sort excels on nearly-sorted data.

Worked Example 1: Merge Sort on Array

Sort $A = [5, 2, 8, 1, 9]$ using merge sort:

Divide: $[5, 2]|[8]|[1, 9]$. Further: $[5]|[2]|[8]|[1]|[9]$.

Merge: Pairs: $[2, 5]$, $[8]$, $[1, 9]$. Three-way: $[2, 5, 8]$, $[1, 9]$. Final: $[1, 2, 5, 8, 9]$.

Worked Example 2: Quick Sort Partition

Partition $A = [3, 7, 2, 8, 1]$ with pivot = 1 (last element):

$i = 0, j = 1$: $A[1] = 7 \not< 1$. Continue.
$j = 2$: $A[2] = 2 \not< 1$. Continue.
$j = 3$: $A[3] = 8 \not< 1$. Continue.
$j = 4$: $A[4] = 1 = 1$. No swap.
Final: $[1, 7, 2, 8, 3]$, pivot at position 0.

Worked Example 3: Radix Sort

Sort integers $[329, 457, 657, 839, 436, 720]$ using radix sort (3 digits):

By ones: $[720, 457, 436, 329, 839, 657]$.
By tens: $[720, 329, 436, 457, 839, 657]$.
By hundreds: $[329, 436, 457, 657, 720, 839]$.

Review Questions

1. Compare bubble, selection, and insertion sort in terms of practical performance.

2. Why is quick sort often faster than merge sort despite worse worst-case complexity?

3. Analyze the time complexity of heap sort and explain why build-heap is $O(n)$.

4. When would you prefer counting sort over comparison-based sorting?

5. How does merge sort maintain stability while sorting?

6. What is the significance of choosing a good pivot in quick sort?

7. Compare radix sort and merge sort for sorting 1 million 32-bit integers.

8. How would you adapt insertion sort to take advantage of nearly-sorted data?

Module 3: Algorithm Design Paradigms

3.1 Divide and Conquer

Divide and conquer solves problems by dividing into subproblems, solving recursively, and combining solutions.

Binary search: Given sorted array, find target. Divide by comparing target to middle element, eliminating half. Recurrence: $T(n) = T(n/2) + O(1)$, solving to $O(\log n)$.

Strassen matrix multiplication: Standard multiplication is $O(n^3)$. Strassen divides matrices into quadrants, performs 7 multiplications and combines results, achieving $O(n^{\log_2 7}) \approx O(n^{2.807})$.

The key is that subproblems are independent and combination is efficient.

3.2 Dynamic Programming

Dynamic programming solves problems with overlapping subproblems and optimal substructure by storing solutions to avoid recomputation.

Overlapping subproblems: Same subproblem appears multiple times.

Optimal substructure: Optimal solution uses optimal solutions to subproblems.

0/1 Knapsack Problem

Given items with weights and values, select subset maximizing value with weight limit W.

Define $dp[i][w]$ = maximum value using first i items with weight limit w.

Recurrence: $dp[i][w] = \max(dp[i-1][w], dp[i-1][w-weight[i]] + value[i])$.

Time: $O(nW)$. Space: $O(nW)$ or $O(W)$ with rolling array.

Worked example: Items: $(weight, value) = [(2,3),(3,4),(4,5),(5,6)]$. $W = 5$.

dp table (rows = items, columns = weights):

Item	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	3	3	3	3
2	0	0	3	4	4	7
3	0	0	3	4	5	7
4	0	0	3	4	5	7

Table 3.1: 0/1 Knapsack DP Table ($W = 5$, 4 items)

Maximum value is $dp[4][5] = 7$ (items 1 and 2).

Longest Common Subsequence

Given strings X and Y, find longest sequence of characters appearing in both in order.

Define $lcs[i][j]$ = length of LCS of $X[1..i]$ and $Y[1..j]$.

Recurrence:

$$lcs[i][j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ lcs[i-1][j-1] + 1 & \text{if } X[i] = Y[j] \\ \max(lcs[i-1][j], lcs[i][j-1]) & \text{otherwise} \end{cases} \tag{3.1}$$

Time and space: $O(mn)$ where $m = |X|, n = |Y|$.

Example: $X =$ "ACGTAC", $Y =$ "TACGTAG". LCS = "ACTAC" or "CGTAC" (length 5).

Matrix Chain Multiplication

Given matrices $M_1, M_2, \ldots, M_n$, find optimal parenthesization minimizing scalar multiplications.

Number of multiplications depends on order. $((M_1M_2)M_3)$ differs from $(M_1(M_2M_3))$.

Define $m[i][j]$ = minimum multiplications to compute $M_i \cdots M_j$.

Recurrence: $m[i][j] = \min_{i \leq k < j}(m[i][k]+m[k+1][j]+rows_i \cdot cols_k \cdot cols_j)$.

Time: $O(n^3)$. Reconstruct optimal parenthesization by storing split point k.

3.3 Greedy Algorithms

Greedy algorithms make locally optimal choices, hoping to achieve globally optimal solutions. They work when:

Greedy choice property: Locally optimal choice leads to globally optimal solution.

Optimal substructure: Optimal solution contains optimal solutions to subproblems.

Activity Selection

Given activities with start/end times, select maximum non-overlapping activities.

Greedy: Always select activity ending earliest. Proof: If a is earliest-ending, then solution with a plus solution to remaining is optimal.

Time: $O(n \log n)$ (after sorting by end time).

Huffman Coding

Build optimal prefix-free code for compression. Greedy: Repeatedly combine two nodes with smallest frequencies, building bottom-up tree.

Proof: Swapping subtrees doesn't reduce cost, and greedy locally minimizes cost at each merge.

3.4 Backtracking

Backtracking explores solution space, pruning branches that can't yield solutions.

N-Queens Problem

Place n queens on $n \times n$ board so no two attack. Backtracking: Place queen row-by-row, check conflicts, backtrack if impossible.

Time: $O(n!)$ worst case, but pruning is effective. Space: $O(n)$ recursion depth.

3.5 Branch and Bound

Branch and bound solves optimization by building search tree with bounds, pruning branches exceeding best found solution.

Traveling Salesman Problem

Find shortest tour visiting all cities. Branch: Extend partial tour. Bound: Lower bound (e.g., minimum spanning tree + lower bound on remaining path). Prune if bound exceeds best solution.

This concludes module 3 on algorithm design paradigms. Let me continue with Module 4 in a continuation...

Review Questions

1. Explain divide and conquer strategy and name three algorithms using it.

2. Define overlapping subproblems and optimal substructure with examples.

3. Derive the recurrence relation for 0/1 knapsack and give time/space complexity.

4. How does longest common subsequence differ from longest increasing subsequence?
5. Compare greedy algorithms and dynamic programming in terms of applicability and efficiency.
6. When is backtracking preferable to other paradigms?
7. Explain the greedy choice property with activity selection example.
8. How do bounds improve branch and bound efficiency?

Module 4: Graph Algorithms and NP-Completeness

4.1 Graph Representations

Graphs model relationships. Representations affect algorithm efficiency.

Adjacency matrix: $n \times n$ matrix A where $A[i][j] = 1$ if edge (i, j) exists. Space: $O(n^2)$. Edge check: $O(1)$. Enumerating neighbors: $O(n)$.

Adjacency list: Array of lists, each storing vertex neighbors. Space: $O(n + m)$ where m is edges. Edge check: $O(\text{degree})$. Enumerating neighbors: $O(\text{degree})$.

Choose based on graph density: Matrix for dense ($m \approx n^2$), list for sparse ($m \ll n^2$).

4.2 Breadth-First Search (BFS)

BFS explores vertices level-by-level from source.

```
Algorithm BFS(G, source)
  visited[source] = true
  queue = {source}
  while queue not empty do
    v = dequeue(queue)
    for each neighbor u of v do
      if not visited[u] then
        visited[u] = true
        enqueue(queue, u)
```

Time: $O(V + E)$. Space: $O(V)$. Properties: Finds shortest paths in unweighted graphs. Visits vertices in breadth-first order.

4.3 Depth-First Search (DFS)

DFS explores as far as possible along each branch before backtracking.

```
Algorithm DFS(G, v)
  visited[v] = true
  for each neighbor u of v do
    if not visited[u] then
      DFS(G, u)
```

Time: $O(V + E)$. Space: $O(V)$ recursion. Properties: Topological sort, strongly connected components, cycle detection.

4.4 Dijkstra's Shortest Path

Finds shortest paths from source to all vertices in weighted graphs (non-negative weights).

```
Algorithm Dijkstra(G, source)
  dist[source] = 0; all others = infinity
  pq = {source with dist 0}
  while pq not empty do
    v = extract-min(pq)
    for each neighbor u of v do
      if dist[v] + weight(v,u) < dist[u] then
        dist[u] = dist[v] + weight(v,u)
        decrease-key(pq, u, dist[u])
```

Time: $O(V^2)$ with array, $O((V + E) \log V)$ with binary heap. Space: $O(V)$. Greedy: Always extends shortest known path.

4.5 Bellman-Ford Algorithm

Finds shortest paths from source, allowing negative weights (but no negative cycles).

```
Algorithm BellmanFord(G, source)
  dist[source] = 0; all others = infinity
  for i = 1 to V-1 do
    for each edge (u,v) with weight w do
      if dist[u] + w < dist[v] then
        dist[v] = dist[u] + w
  for each edge (u,v) with weight w do
    if dist[u] + w < dist[v] then
      report negative cycle
```

Time: $O(VE)$. Space: $O(V)$. DP approach: Relaxes edges $V - 1$ times.

4.6 All-Pairs Shortest Paths: Floyd-Warshall

Computes shortest paths between all pairs.

```
Algorithm FloydWarshall(G)
  dp[i][j] = weight(i,j)
  for k = 1 to V do
    for i = 1 to V do
      for j = 1 to V do
        dp[i][j] = min(dp[i][j], dp[i][k] + dp[k][j])
```

Time: $O(V^3)$. Space: $O(V^2)$. DP approach: $dp[i][j]$ after k is shortest path using vertices $1..k$ as intermediates.

4.7 Minimum Spanning Tree

MST connects all vertices with minimum total weight edges. Two greedy algorithms:

Kruskal: Sort edges, add edge if it doesn't create cycle (using Union-Find).

Time: $O(E \log E + E\alpha(V))$ where α is inverse Ackermann (nearly constant).

Prim: Start from vertex, repeatedly add minimum edge connecting tree to new vertex.

Time: $O(V^2)$ with array, $O(E \log V)$ with heap.

Both greedy approaches work because MST property: Adding minimum edge across cut always yields MST.

4.8 NP-Completeness

P (Polynomial): Problems decidable in polynomial time. Examples: sorting, shortest path, 2-SAT.

NP (Nondeterministic Polynomial): Problems verifiable in polynomial time. Example: 3-SAT (given assignment, verify in $O(n)$).

NP-hard: Problems at least as hard as hardest NP problem. Reducing known NP-hard problem to problem X shows X is NP-hard.

NP-complete: NP-hard and in NP. Example: 3-SAT, Clique, Vertex Cover, TSP (decision version).

Reduction: Polynomial-time reduction from A to B shows if B solvable in poly time, so is A. Contrapositive: if A hard, so is B.

Example reduction: 3-SAT $\leq$ Clique. Given 3-SAT instance, construct graph where vertices are literals, edges connect compatible literals. Clique of size k corresponds to satisfying assignment of k clauses.

4.9 Approximation Algorithms

NP-complete problems may not have polynomial-time exact algorithms. Approximation algorithms find near-optimal solutions.

Vertex Cover: Select minimum vertices covering all edges. Greedy 2-approximation: Repeatedly pick edge, add both endpoints, remove incident edges. Achieves at most $2\times$ optimal.

Proof: Optimal covers m edges; greedy picks $2k$ vertices covering at least k distinct edges (uncovered by any vertex), so optimal

$\geq k$, greedy $\leq 2k$.

Approximation ratio of r means algorithm always achieves $\leq r\times$ optimal (minimization) or $\geq 1/r\times$ optimal (maximization).

Worked Example 1: Dijkstra's Algorithm

Graph with edges: $(s, a, 2), (s, b, 4), (a, b, 1), (a, c, 5), (b, c, 2)$.

Initial: dist = $[0, \infty, \infty, \infty]$.

Step 1: Extract s (dist 0). Relax: $a \leftarrow 2, b \leftarrow 4$.

Step 2: Extract a (dist 2). Relax: $b \leftarrow \min(4, 2 + 1) = 3, c \leftarrow 7$.

Step 3: Extract b (dist 3). Relax: $c \leftarrow \min(7, 3 + 2) = 5$.

Step 4: Extract c (dist 5).

Final: dist = $[0, 2, 3, 5]$.

Worked Example 2: Kruskal's MST

Edges: $(a, b, 1), (a, c, 2), (b, c, 3), (b, d, 4), (c, d, 5)$.

Sort: $(a, b, 1), (a, c, 2), (b, c, 3), (b, d, 4), (c, d, 5)$.

Add $(a, b, 1)$. Add $(a, c, 2)$. Skip $(b, c, 3)$ (creates cycle). Add $(b, d, 4)$.

MST edges: $(a, b, 1), (a, c, 2), (b, d, 4)$. Total weight: 7.

Worked Example 3: 3-SAT Reduction to Clique

3-SAT clause: $(x_1 \vee \bar{x}_2 \vee x_3)$.

Graph vertices: For each clause, three vertices (one per literal). Edge between compatible literals (not contradictory).

$(x_1, \bar{x}_2), (x_1, x_3), (\bar{x}_2, x_3)$ in different clauses: All compatible.

Clique of size 3 represents satisfying assignment.

Review Questions

1. Compare adjacency matrix and adjacency list representations.
2. What is the time complexity of BFS and DFS, and when do you use each?

3. How does Dijkstra differ from Bellman-Ford in handling negative weights?

4. Explain the DP approach in Floyd-Warshall algorithm.

5. Define NP-completeness and give three NP-complete problems.

6. What is a polynomial-time reduction and how does it establish hardness?

7. How does a greedy 2-approximation for Vertex Cover prove near-optimality?

8. Compare the complexity of finding MST with Kruskal vs. Prim.

Appendix: Extended Topics

Advanced Sorting: Hybrid Approaches

Introsort (introspective sort) combines quicksort, heapsort, and insertion sort. Begin with quicksort, monitor recursion depth. If depth exceeds $2 \log n$, switch to heapsort. On small subarrays, switch to insertion sort. Achieves $O(n \log n)$ worst-case while being faster than pure quicksort on average.

Timsort (Python's standard sort) divides array into runs, sorts with insertion sort, merges with merge sort optimized for partially-sorted data. Excels on real-world data with existing order.

External Sorting

For data larger than memory, external sorting minimizes disk I/O. Merge external runs: Read blocks into memory, sort, write back. Merge sorted blocks, minimizing pass count.

Time: $O(n \log_M n)$ where M is available memory and n is data size, accounting for block I/O cost.

String Matching Algorithms

Given text and pattern, find all occurrences.

Naive: $O(nm)$ by comparing at each position.

Knuth-Morris-Pratt (KMP): Precompute failure function, avoid redundant comparisons. $O(n + m)$.

Rabin-Karp: Hash pattern and text substrings, compare hashes. $O(n + m)$ expected with good hash function.

Boyer-Moore: Start matching from pattern end, use mismatch information to skip positions. $O(n/m)$ best case, $O(nm)$ worst case. Fastest in practice.

Geometric Algorithms

Closest pair: Find two points with minimum distance. Divide and conquer: $O(n \log n)$.

Convex hull: Find outermost points. Graham scan: Sort by angle, maintain counterclockwise turn. $O(n \log n)$.

Line segment intersection: Sweep line algorithm processes events, maintains active segments. $O(n \log n + k)$ where k is intersections.

Randomized Algorithms

Randomized quicksort: Choose random pivot. Expected $O(n \log n)$ time, avoiding worst-case $O(n^2)$ with adversarial inputs.

Las Vegas: Correct answer always, randomized time. Sherwood: Fixed time, probabilistic correctness.

Randomized algorithms often simpler and faster than deterministic counterparts.

Approximation Algorithms: Further Examples

TSP: 2-approximation using MST. Visit all MST vertices via DFS, shortcut to nearest unvisited. Total distance $\leq$ 2× MST $\leq$ 2× optimal TSP.

Set Cover: Greedy cover selecting set covering most uncovered elements. $O(\log n)$ approximation.

Knapsack: Dynamic programming (pseudo-polynomial $O(nW)$) is exact. FPTAS (fully polynomial-time approximation scheme) provides $(1 + \epsilon)$ approximation in poly$(n, 1/\epsilon)$ time.

Memoisiation vs. Tabulation

Dynamic programming has two implementations:

Memoization (top-down): Recursively compute subproblems, cache results. Natural but may have large stack depth.

Tabulation (bottom-up): Fill table iteratively in dependency order. More efficient, requires identifying order.

Choose based on clarity, problem structure, and memory constraints.

Practical Algorithm Selection

When choosing algorithms:

Consider constraints: Time limit, memory limit, input size.

Start simple: Use naive algorithms first, optimize if needed.

Profile: Measure actual performance. Asymptotic complexity doesn't capture constants.

Use libraries: Most languages provide optimized built-in algorithms. Avoid reinventing.

Understand tradeoffs: Fastest algorithm may use most memory. Balance based on constraints.

Test edge cases: Algorithms may behave differently on small inputs, sorted data, duplicates.

Document assumptions: Clarify whether algorithm requires sorted input, handles duplicates, etc.

This concludes our comprehensive coverage of algorithm analysis, design paradigms, and specific algorithms across domains.

Additional Topics in Algorithmic Design

Linear Programming and Optimization

Linear programming solves optimization problems with linear objective and constraints. Simplex method finds optimal solution in polynomial time empirically, though worst-case is exponential. Interior-point methods have polynomial worst-case.

Example: Maximize $3x + 2y$ subject to $x + y \leq 4$, $x \leq 2$, $y \leq 3$, $x, y \geq 0$.

Many problems reduce to LP: Shortest path (with negative weights when relaxed), maximum flow, minimum cost flow.

Network Flow Algorithms

Maximum flow: Given graph with capacities, find maximum flow from source to sink.

Ford-Fulkerson: While augmenting path exists, increase flow by bottleneck capacity. Time depends on capacity magnitudes or edge count.

Edmonds-Karp (BFS-based Ford-Fulkerson): $O(VE^2)$ with integer capacities. Push-relabel algorithm: $O(V^3)$ or $O(V^2\sqrt{E})$ with gap heuristics.

Applications: Bipartite matching (maximum matching via max flow), airline scheduling, circulation problems.

Convex Optimization

Many problems have convex structure enabling efficient solution.

Gradient descent: Move in direction of steepest descent. Converges to global minimum for convex functions.

Newton's method: Uses second-order information (Hessian) for faster convergence.

Coordinate descent: Optimize one variable while fixing others, cycle through variables.

These methods underlie modern machine learning optimization (SGD, Adam, etc.).

Randomized Algorithms: Probabilistic Methods

Probabilistic method: Prove existence of object without constructing it. Show that random object has desired property with positive probability.

Example: Las Vegas algorithms may fail (randomized) but always correct (if succeed). Sherwood algorithms always succeed with randomized time.

Randomized quicksort, randomized mincut, etc. illustrate power of randomization in avoiding worst-case.

Parallel and Distributed Algorithms

Parallel algorithms utilize multiple processors. Model: PRAM (parallel RAM) with p processors sharing memory. Analysis: Depth (critical path length) and work (total operations).

Example: Parallel merge uses p processors to merge two sorted arrays in $O(\log^2 n)$ depth with $O(n \log n)$ work. Sequential merge is $O(n)$.

Distributed algorithms execute on multiple machines with message passing. Network graph models machines and communication links. Challenges: Asynchronous execution, node/link failures, message delays.

Consensus problem: Nodes must agree on value despite some failures. Byzantine fault tolerance algorithms withstand malicious nodes.

Computational Geometry Algorithms

Point location: Given planar subdivision and query point, find containing region. Trapezoidal decomposition enables $O(\log n)$ query with $O(n)$ space.

Voronoi diagram: Partition plane into regions nearest to each site. Applications: Largest empty circle, nearest neighbor queries. Construction: Fortune's algorithm in $O(n \log n)$.

Delaunay triangulation: Dual of Voronoi. Properties: Maximizes minimum angle (avoids thin triangles). Applications: mesh generation, terrain modeling.

Convex hull: Graham scan, Gift wrapping, Divide and conquer. All $O(n \log n)$.

Pattern Matching and Regular Expressions

Finite automata approach: Build automaton matching pattern, run in linear time.

Nondeterministic finite automaton (NFA): Multiple states with same input symbol. Accepts string if any path reaches accept state.

Deterministic finite automaton (DFA): Single next state per input symbol. Can be built from NFA via subset construction (possibly exponential states).

Regex engines: Often use NFA for simplicity or DFA for speed (depending on regex flavor).

Thompson's construction: Builds NFA from regex in $O(m)$ time.

Competitive Analysis

Competitive analysis measures performance of online algorithms compared to offline (all input known in advance) algorithms.

k-competitive: Online algorithm costs at most k times offline optimal.

Move-to-front heuristic for list access: Pay cost when accessing element, move to front. Competitive ratio: 2 (any online algorithm is $\Omega(n)$-competitive in worst-case).

Paging: Cache replacement algorithms. Offline: BÉLADY's algorithm (page fault offline optimal). Online: LRU, FIFO. LRU is k-competitive for k cache pages.

Ski rental problem: Rent vs. buy. Rent costs \$1/day, buy costs \$k. Optimal offline: Buy if slope $\geq \log k$. Rent fraction: $\log k$-competitive.

Primal-Dual and LP-Rounding

Primal-dual algorithms solve problems by maintaining feasible primal and dual solutions, improving both toward optimality.

Set Cover: LP relaxation allows fractional cover. Dual LP: Assign weights to elements. Primal-dual algorithm: Increase weights, add sets covering heavy elements. Achieves $O(\log n)$ approximation.

LP-rounding: Solve relaxation, round to integer solution. Vertex Cover: LP relaxation is $\geq n/2$ vertices. Round vertex v if LP value $\geq 1/2$. Achieves 2-approximation.

Randomized rounding: Include set with probability equal to LP solution value. Expected cost in poly time, high probability bounds via concentration inequalities.

Derandomization

Convert randomized algorithms to deterministic via method of conditional expectations or other techniques.

Example: MAX-SAT randomized algorithm: Assign variables randomly. Expected satisfied clauses: $\geq 7/8 \cdot m$ (for clauses with ≥ 1 literal). Derandomize: Decide each variable greedily (whichever value satisfies more clauses).

Derandomization often incurs efficiency loss (more computation, sometimes worse asymptotic complexity).

Approximation Schemes

PTAS (polynomial-time approximation scheme): $(1+\epsilon)$-approximation in time poly$(n, 1/\epsilon)$. Example: Euclidean TSP.

FPTAS (fully polynomial): Time poly$(n, 1/\epsilon)$. Example: 0/1 Knapsack with value scaling.

APTAS: Approximation poly$(1/\epsilon)$-factor longer. Example: General TSP.

Hardness of approximation: Some problems resist approximation. Gap results show approximation harder than exact solution.

Practical Algorithm Engineering

Constant factors matter: $O(n)$ algorithm 10x slower than $O(n \log n)$ algorithm for small inputs. Profile before optimizing.

Cache efficiency: Memory layout affects performance. Algorithms with good cache locality (sequential access) run 10x-100x faster than random access.

Vector instructions: Modern CPUs support SIMD (single instruction, multiple data). Algorithms leveraging this (e.g., bitwise operations for subset enumeration) gain 4-16x speedup.

Parallel implementations: Utilize multiple cores for 2-8x speedup (depending on algorithm parallelizability and core count).

Approximation in practice: Exact algorithms often too slow for large instances. Use approximations, local search, or heuristics instead.

Example: Traveling Salesman: Exact algorithms intractable for $n > 20$. 2-opt local search: Fast, achieves $\approx 95\%$ of optimal on average instances.

This extensive coverage provides tools, theoretical foundations, and practical insights for algorithm design and analysis across diverse problem domains.

Case Studies: Algorithms in Real-World Systems

Google Search PageRank

PageRank algorithm ranks web pages by importance. Treat web as directed graph: vertices = pages, edges = links. PageRank of page p:

$$PR(p) = \frac{1-d}{N} + d \sum_{q \to p} \frac{PR(q)}{L(q)} \tag{4.1}$$

where $d = 0.85$ (damping factor), N = total pages, $L(q)$ = number of outgoing links from q, sum over pages q linking to p.

Iterative computation until convergence: $O(E \cdot \text{iterations})$ where E is edges. Real web has billions of pages, trillions of edges, but sparse structure enables efficient computation.

Netflix Recommendation: Collaborative Filtering

Netflix recommends movies based on user ratings. Collaborative filtering: Similar users like similar movies.

Build user-item rating matrix: Users $\times$ Movies. Use nearest neighbors or matrix factorization (SVD, NMF) to predict missing ratings.

Matrix factorization: Approximate matrix as product of lower-rank matrices. User matrix: $n \times k$, Movie matrix: $m \times k$. Learned via alternating least squares.

Time complexity for m users, n items, k factors: $O(kmn \cdot \text{iterations})$. Practical systems use millions of users, millions of items, $k \approx 100$.

Optimization: Stochastic gradient descent, distributed computation (MapReduce, Spark), approximations.

LZ77 Compression (zip, gzip, zlib)

LZ77 exploits repetition in data. Sliding window: Maintain recent data in buffer. For current position, find longest match in buffer, output (offset, length) pair instead of literal bytes.

Time complexity: Naïve $O(n \cdot \text{window_size})$. Optimized with hash tables: $O(n)$ or $O(n \log n)$.

Compression ratio: 2-10x for text, less for binary or compressed data.

Burrows-Wheeler Transform and Suffix Arrays

Burrows-Wheeler Transform (BWT): Rearrange string for better compressibility. Rotate string all ways, sort, output last column. Transform is reversible.

Example: ”banana” rotations: [banana, ananab, nanaba, anaban, nabana, aban]. Sorted: [aban, ananab, anaban, banana, nabana, nanaba]. Last column: ”annbaa”.

BWT clustering equal characters, enabling better compression (run-length encoding, move-to-front, Huffman).

Suffix array: Array of sorted suffixes. Enables fast string matching, pattern search. Construction: $O(n \log n)$ (sorting) or $O(n)$ (specialized algorithms).

Multimedia Codecs: JPEG, MPEG, H.264

JPEG compression: Divide image into 8×8 blocks, apply discrete cosine transform (DCT), quantize high-frequency components (imperceptible), entropy encode.

MPEG video: Exploits spatial (within frame) and temporal (between frames) redundancy. Key frames (I-frames) are fully encoded. Predicted frames (P-frames) code differences from previous. Bidirectional frames (B-frames) code differences from previous and future.

H.264 advanced codec: Hierarchical block decomposition, multiple reference frames, improved motion compensation. Achieves 2-3x better compression than MPEG2.

GPS Navigation and Route Planning

Route planning in road networks: Graph with intersections (vertices), roads (edges), distances (weights). Goal: Find shortest path from source to destination.

Naïve Dijkstra: $O(V^2)$ or $O(E \log V)$, too slow for million-vertex graphs.

Optimization: Bidirectional Dijkstra (search from both ends, meet in middle). Speeds up 2x.

Better: Hierarchical graphs. Precompute contracted hierarchies: Remove non-highway vertices, preserving shortest paths. Search on hierarchy: Exponentially fewer vertices.

Contraction hierarchies enable $O(\log V)$ query time on road networks.

Alternative: Landmark-based lower bounds. Precompute distances to landmarks, use as heuristic in A* algorithm. Dramatically reduces search space.

Spell Checking and Edit Distance

Edit distance (Levenshtein): Minimum operations (insert, delete, substitute) transforming string A to B. DP solution: $O(|A| \cdot |B|)$

time and space.

Spell checking: For misspelled word, find dictionary words with small edit distance. Optimize with tries and early termination (prune branches exceeding threshold).

Fast approximations: Restrict substitutions, use deletion only, consider only single operations.

Machine Learning: Gradient-Based Optimization

Neural networks optimized via stochastic gradient descent variants (SGD, momentum, Adam).

Forward pass: Compute predictions. Backward pass (backpropagation): Compute gradients via chain rule.

Time complexity: $O(n \cdot \text{layers} \cdot \text{units}^2)$ per iteration.

Optimization: Batching (amortize computation), mixed precision (float16 + float32), distributed training (data parallelism), sparse computation (GPU kernels for specific operations).

Training 175B-parameter models (GPT-3) requires thousands of GPUs, weeks of computation, billions of dollars.

Cryptocurrency and Blockchain: Proof of Work

Bitcoin consensus: Nodes compete to solve computational puzzle (hash function inversion), finding hash with specific leading zeros. Cost: Work equivalent. Benefit: Honest nodes dominate (need 51% hash power to attack).

Mining: Iterate nonce in block until hash satisfies condition. Expected work: $2^{\text{difficulty}}$ hash computations.

Difficulty adjusts every 2 weeks to maintain 10-minute block time.

Memory-hard puzzles (Ethash, used by Ethereum): Require memory and compute, resisting specialized hardware (ASICs).

Cryptography: Public-Key Encryption

RSA: Factoring $n = p \cdot q$ (large primes) is hard. Encryption/decryption: Modular exponentiation $O(\log n)$ multiplications (via binary exponentiation).

Elliptic Curve: Discrete logarithm on elliptic curves is hard. Shorter keys (256-bit ECC $\approx$ 2048-bit RSA in security).

Hash functions (SHA-256, etc.): One-way functions, collision-resistant. Used in digital signatures, blockchain, password hashing.

This case-study section demonstrates algorithm concepts in real systems, bridging theory and practice.

Advanced Topics and Future Directions

Streaming Algorithms

Data arrives as stream, cannot reprocess or store fully. Algorithms must process data once or few times with limited memory.

Count-min sketch: Approximate frequency of items. Hash items into array, increment counts. Query: Return minimum count across hashes. Uses $O(\epsilon^{-1} \log^{-1} \delta)$ space for (ϵ, δ) approximation.

Distinct count (cardinality): HyperLogLog algorithm estimates number of distinct elements in $O(\log \log n)$ space with $O(1)$ time per element.

Median finding: Reservoir sampling or quantile sketches. Sample representative subset, compute approximate quantiles.

Applications: Network traffic analysis (count flows, traffic per source), database query execution (estimate output size), anomaly detection.

Quantum Computing Algorithms

Quantum algorithms exploit superposition and entanglement for speedups.

Shor's algorithm: Factor integers in $O(n^3)$ (exponentially faster than classical $O(2^n)$). Threatens RSA encryption.

Grover's search: Unsorted database search in $O(\sqrt{n})$ (quadratic speedup over classical $O(n)$).

Quantum advantage: Limited to specific problem structures. General-purpose speedup unlikely; most algorithms see at most polynomial speedup.

Current state: 50-500 qubit prototypes. Practical quantum computers years away.

Machine Learning Model Optimization

Model compression: Reduce model size without sacrificing accuracy.

Pruning: Remove unimportant weights/neurons. 95% of weights in trained networks are often redundant.

Quantization: Use lower precision (8-bit integer vs. 32-bit float). Reduces size 4x, minimal accuracy loss.

Knowledge distillation: Train small model to imitate large model. Small model achieves large model's accuracy.

Neural architecture search: Automatically find small, efficient architectures. Discover models with 100x fewer parameters at equal accuracy.

Federated learning: Train on distributed devices without centralizing data. Addresses privacy, communication bottlenecks.

Competitive Programming Techniques

Competitive programmers solve hard problems under time pressure. Techniques:

Meet-in-the-middle: Divide problem into two halves, enumerate solutions, match. Reduces 2^n to $2^{n/2}$.

Bitmasking: Encode set as bitmask, iterate via bitmask enumeration. $O(3^n)$ algorithms for subset problems.

Square root decomposition: Divide input into $\sqrt{n}$ blocks. Trade time vs. space: $O(\sqrt{n})$ per query and update instead of $O(1)$ or $O(n)$.

Segment trees, Fenwick trees: Range query and update in $O(\log n)$.

Heavy-light decomposition: Decompose tree into heavy and light paths, enable fast path/subtree queries.

Convex hull trick: For linear function optimization with specific structure, reduce to convex hull queries.

These techniques extend algorithmic toolkit for extreme problem-solving environments.

Algorithm Correctness Verification

Formal verification: Prove algorithm correctness via mathematical proof.

Induction: Prove loop invariants, conclude correctness.

Worst-case analysis: Analyze worst-case time complexity.

Probabilistic proofs: For randomized algorithms, prove expected behavior.

Model checking: Automatically verify finite-state systems against specifications.

Theorem provers (Coq, Isabelle, Lean): Mechanically verify proofs, reducing human error.

Practical verification: Extensive testing, code review, assertions, invariant checking at runtime.

Interdisciplinary Applications

Algorithms appear in diverse domains:

Computational biology: DNA sequence alignment (edit distance), gene finding (Hidden Markov Models), protein folding (NP-hard but heuristics effective).

Computational chemistry: Molecular simulation, drug discovery via conformational sampling.

Climate modeling: Solve massive differential equations on supercomputers. Algorithms for parallelization critical.

Finance: Options pricing (Black-Scholes via PDEs), portfolio optimization (convex optimization).

Game theory: Computing Nash equilibrium, minimax game trees.

Operations research: Vehicle routing, facility location, resource allocation (integer programming).

Each domain applies core algorithmic techniques to domain-specific problems.

This extensive appendix provides advanced topics, real-world applications, and future directions in algorithmic research and practice.

Summary of Algorithm Design Principles

Throughout this book, we've developed several key principles:

Understand the problem: Define input/output clearly, identify constraints, determine if problem is NP-hard.

Analyze complexity: Best, average, worst-case. Asymptotic and concrete complexity. Space and time tradeoffs.

Choose appropriate paradigm: Divide and conquer for recursive structure. Dynamic programming for overlapping subproblems. Greedy for local optimality. Backtracking for exhaustive search.

Implement efficiently: Use appropriate data structures. Constant factors matter. Profile and optimize. Use existing libraries when available.

Verify correctness: Prove correctness via invariants or induction. Test on diverse cases. Consider edge cases.

Evaluate pragmatically: Measure actual performance. Compare to baselines. Consider deployment constraints.

Resources for Continued Learning

Classical algorithms texts: CLRS (Cormen, Leiserson, Rivest, Stein), Knuth volumes, Sedgewick and Wayne.

Online resources: MIT OpenCourseWare (6.046J), Stanford Algorithms courses, TopCoder, CodeForces for competitive programming.

Research conferences: ACM STOC, IEEE FOCS, SIAM SODA for theoretical advances.

Journals: JACM, Algorithmica, Journal of Algorithms for peer-reviewed publications.

Code repositories: GitHub for reference implementations of algorithms.

Reflection on Algorithmic Thinking

Algorithms are humanity's way of explicitly describing solutions. They are precise, unambiguous, and executable. Studying algorithms develops problem-solving skills applicable beyond computer science.

Algorithmic thinking: Break problems into subproblems. Identify patterns. Leverage structure. Measure efficiency. Optimize systematically.

These skills apply in mathematics, physics, engineering, economics, medicine, and everyday life.

As computational problems grow larger and computational resources more powerful, algorithmic innovation remains essential. Faster algorithms directly enable new applications: processing gigabytes of data, training billion-parameter models, simulating complex systems.

We encourage you to engage deeply with algorithms: implement them, modify them, combine them, invent new ones. The field evolves as practitioners develop novel techniques for emerging problems.

Algorithm design is both art and science: art in recognizing problem structure and crafting elegant solutions, science in rigorous analysis and verification. Master both aspects to become an excellent algorithm designer.

Index of Important Concepts

Throughout the book, we've introduced many concepts. This index provides quick references:

Recurrence relations: $T(n) = aT(n/b) + f(n)$ solved via substitution, iteration, or Master Theorem.

Sorting: Bubble ($O(n^2)$), Selection ($O(n^2)$), Insertion ($O(n)$ to $O(n^2)$), Merge ($O(n \log n)$), Quick ($O(n^2)$ worst, $O(n \log n)$ average), Heap ($O(n \log n)$), Counting ($O(n+k)$), Radix ($O(d(n+k))$).

Data structures: Array, Linked List, Stack, Queue, Heap, Hash Table, Binary Search Tree, AVL Tree, Segment Tree, Fenwick Tree, Union-Find, Trie.

Graph algorithms: BFS ($O(V+E)$), DFS ($O(V+E)$), Topological Sort, SCC via DFS, Dijkstra ($O(E \log V)$), Bellman-Ford ($O(VE)$), Floyd-Warshall ($O(V^3)$), Kruskal ($O(E \log E)$), Prim ($O(E \log V)$).

String algorithms: Naive match ($O(nm)$), KMP ($O(n+m)$), Rabin-Karp ($O(n+m)$ expected), Boyer-Moore ($O(nm)$ worst, $O(n/m)$ best).

Paradigms: Divide and Conquer, Dynamic Programming, Greedy, Backtracking, Branch and Bound.

Complexity classes: P (polynomial), NP (verifiable in polynomial), NP-hard, NP-complete.

Approximation: Vertex Cover (2-approx), Set Cover ($O(\log n)$-approx), TSP (1.5-approx via Christofides).

Problem-Solving Checklists

When facing a new problem:

Problem Analysis Checklist

- Clarify input/output format and constraints.

- Estimate input size (small $n \leq 10^6$ allows $O(n \log n)$; large $n \leq 10^8$ requires $O(n)$).
- Research if problem is known (likely solved in literature).
- Check if problem is NP-hard (then approximate, heuristic, or parameterized).
- Identify problem structure: graph, string, arithmetic, geometric, combinatorial.

Algorithm Design Checklist

- Try simple solutions first (brute force, greedy, obvious structure).
- Identify subproblems and dependency structure (suggests divide and conquer or DP).
- Consider data structure choice (array vs. heap vs. hash table).
- Analyze time and space complexity.
- Verify correctness on small examples and edge cases.
- Implement efficiently (avoid redundant computation, use appropriate library).
- Test on diverse inputs: random, adversarial, edge cases.

Optimization Checklist

- Profile to identify bottlenecks (don't optimize prematurely).
- Reduce constant factors (use fast operations, avoid memory allocations).
- Improve asymptotic complexity (algorithmic improvement over constant factors).

- Parallelize if possible (multiple threads, distributed computation).
- Use caching and memoization (trade memory for speed).
- Simplify problem: restrictive input, approximation, greedy.

This practical guidance summarizes our journey through algorithms, from foundational analysis to advanced applications and optimization techniques.

Glossary

Amortized Analysis: Average cost per operation over sequence of operations, accounting for expensive operations amortized over many cheap ones.

Big-O: Asymptotic upper bound on function growth rate, ignoring constants and lower-order terms.

Branch and Bound: Optimization technique building search tree with pruning based on bounds.

DAG: Directed Acyclic Graph, enabling topological sort and dynamic programming.

Divide and Conquer: Paradigm solving problem by dividing into subproblems, solving recursively, combining.

Dynamic Programming: Paradigm solving problems with overlapping subproblems by storing solutions.

Greedy: Paradigm making locally optimal choices hoping for global optimality.

Hash Table: Data structure enabling $O(1)$ average-case search, insertion, deletion.

Heap: Binary tree maintaining heap property enabling priority queue operations in $O(\log n)$.

In-place: Algorithm modifying input directly without extra storage proportional to input size.

MST: Minimum Spanning Tree connecting all vertices with minimum total edge weight.

NP-complete: In NP and NP-hard; all NP problems reduce to it.

NP-hard: At least as hard as NP problems; harder than all NP-complete problems.

Reduction: Polynomial-time algorithm transforming instance of problem A to problem B.

Recurrence: Recursive equation for algorithm's time complexity relation.

Shortest Path: Minimum-weight path between two vertices in weighted graph.

Stable Sort: Sorting preserving relative order of equal elements.

SCC: Strongly Connected Component; maximal set of vertices with paths between all pairs.

Topological Sort: Linear ordering of DAG vertices respecting edge directions.

Union-Find: Disjoint-set data structure enabling fast union and find operations ($O(\alpha(n))$).

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